

API for Beginners - Visual Study Guide

1. What is an API?

An API (Application Programming Interface) allows one application to communicate with another. Example: Uber uses Google Maps APIs to show routes. Flow: User -> App -> API -> Service -> Response.

2. Interfaces and Abstraction

An interface hides complexity. A steering wheel lets you drive a car without understanding the engine. Similarly, APIs hide the internal implementation of software.

3. Local vs Remote APIs

Local APIs run on your machine (Python libraries like NumPy, Pandas). Remote APIs run over the internet (Spotify API, OpenAI API, Weather APIs).

4. How the Web Works

Browser -> HTTP Request -> Server -> HTTP Response -> Browser. Most APIs work using the same request/response model.

5. HTTP Methods

GET = Read data. POST = Create data. PUT = Replace data. PATCH = Update data. DELETE = Remove data.

6. REST APIs

REST treats everything as a resource accessed through URLs. Examples: /users, /albums, /products.

7. CRUD Operations

Create, Read, Update and Delete map directly to HTTP methods.

8. JSON

JSON is the most common API data format. Example: {"name":"Palash","role":"AI Engineer Student"}

9. Status Codes

200 Success, 201 Created, 400 Bad Request, 401 Unauthorized, 404 Not Found, 500 Server Error.

10. Authentication

APIs often require API Keys, OAuth tokens, JWTs or Basic Authentication to identify users.

11. OAuth

OAuth lets users authorize applications without sharing passwords. Spotify uses OAuth extensively.

12. Spotify API Example

Search artists, albums and songs. Learn pagination, authentication, JSON responses and API documentation.

13. Curl

Curl is a command-line tool for testing APIs. Example: `curl https://api.example.com/users`

14. Twilio API Example

Send SMS messages, make phone calls, send WhatsApp messages and media using API requests.

15. Postman and RESTFox

GUI tools used to test APIs, inspect requests and responses, and manage collections.

16. SDKs

SDKs simplify API usage. Instead of building HTTP requests manually, developers call library functions.

17. REST Constraints

Client-Server, Stateless, Cacheable, Uniform Interface, Layered System, Code on Demand.

18. HATEOAS

Advanced REST concept where responses include links to related resources.

19. Project 1 - Weather Dashboard

Use a free weather API. Build a Flask application that accepts a city and displays weather information. Skills learned: HTTP requests, JSON parsing, API keys, Flask.

20. Project 2 - AI News Aggregator

Use NewsAPI and OpenAI API. Fetch AI news articles, summarize them, and display results on a Flask website. Skills learned: Multiple APIs, automation, JSON processing, web development.

21. Interview Questions

What is an API? What is REST? Explain GET vs POST. What is JSON? What is OAuth? What are status codes?

22. Cheat Sheet

API = Communication layer between applications. REST = Architecture. JSON = Data format.
GET=Read POST=Create PUT=Replace PATCH=Update DELETE=Remove.